LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Light – Reflection and Refraction

Formulas & sign convention

	Mirror	Lens
Formula	1/v + 1/u = 1/f	1/v – 1/u = 1/f
Magnification	m = h′/h = –v/u	m = h′/h = v/u
Focal length	f = R/2	Power P = 1/f (f in metres), unit dioptre (D)
Sign of f	Concave –, convex +	Convex +, concave –

New Cartesian convention: distances are measured from the pole or optical centre. Object on the left, so u is always negative. Distances along the incident light are +, against it are –. Heights above the axis are +, below are –. **Negative m = real and inverted. Positive m = virtual and erect.**

Concave mirror: image formation

Object at	Image at	Size	Nature
Infinity	F	Point-sized	Real, inverted
Beyond C	Between F and C	Diminished	Real, inverted
C	C	Same size	Real, inverted
Between C and F	Beyond C	Enlarged	Real, inverted
F	Infinity	Highly enlarged	Real, inverted
Between P and F	Behind the mirror	Enlarged	Virtual, erect

Convex lens: image formation

Object at	Image at	Size	Nature
Infinity	F_2	Point-sized	Real, inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real, inverted
$2F_1$	$2F_2$	Same size	Real, inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real, inverted
F_1	Infinity	Highly enlarged	Real, inverted
Between F_1 and O	Same side as the object	Enlarged	Virtual, erect

Convex mirror and **concave lens** always give a virtual, erect, diminished image (convex mirror: behind the mirror between P and F; concave lens: between F_1 and O).

Refraction

- Light bends because its speed changes. Going into a denser medium (air → glass) it bends **towards** the normal; going into a rarer medium it bends away.
- Laws: incident ray, refracted ray and normal lie in one plane; **sin i / sin r = constant** (Snell's law).
- Refractive index n = c/v (speed in vacuum ÷ speed in medium). $n_{21} = v_1/v_2$. Glass slab: the emergent ray is parallel to the incident ray but shifted sideways.
- Values to remember: water 1.33, kerosene 1.44, crown glass 1.52, diamond 2.42 (optically densest).

Uses. Concave mirror: torches and headlights, shaving and dentists' mirrors, solar furnaces. Convex mirror: rear-view mirrors (erect image, wider view).
Ray-diagram rules: a ray parallel to the axis passes through (or appears to come from) F; a ray through F becomes parallel; a ray through C (mirror) returns along itself; a ray through O (lens) goes straight.

Quick check

1. Power of a lens is –2.5 D. What is its focal length, and what type of lens is it?

2. Where should an object be placed in front of a concave mirror to get a same-size real image?